

Problem Formulation

Governing Equation:

Continuity $\rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

X-Momentum $\rightarrow u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x}$

Y-Momentum $\rightarrow u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial y}$

Solve for (u, v, p)

Reformulation

Recall that,

$$\omega = \nabla \times U = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} \xrightarrow{\text{in } x-y \text{ plane}} \omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

$$u = \frac{\partial \psi}{\partial y} \quad v = -\frac{\partial \psi}{\partial x}$$

$$\therefore \omega = -\frac{\partial}{\partial x} \left(\frac{\partial \psi}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \psi}{\partial x} \right) \rightarrow \nabla^2 \psi = -\omega \quad \boxed{1}$$

$\uparrow$ Laplacian $\rightarrow$ Divergence (gradient)

Differentiation w.r.t y

$$u_y u_x + u u_{xy} + v_y u_y + v u_{yy} = -\frac{1}{\rho} p_{xy} \rightarrow (1)$$

w.r.t x

$$u_x v_x + u v_{xx} + v_x v_y + v u_{yx} = -\frac{1}{\rho} p_{yx} \rightarrow (2)$$

$\therefore$ We are dealing with incompressible $\rightarrow$ no discontinuities ($p_{xy} = p_{yx}$)

Subtract (2) from (1)

$$\therefore u \frac{\partial}{\partial x} (v_x - u_y) + v \frac{\partial}{\partial y} (v_x - u_y) = 0$$

$$\therefore u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} = 0$$

$$\therefore \frac{\partial \omega}{\partial t} + u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} = 0 = \frac{D\omega}{Dt} \quad \boxed{2}$$

Zero $\rightarrow$

$\boxed{2} \rightarrow$ Reads the total time derivative of the vorticity is zero $\rightarrow$ vorticity is conserved along a stream line $\rightarrow \omega = \text{const}$

It doesn't mean everywhere in the domain but rather a number along a stream line and differ from one streamline to another $\rightarrow$ so it's a function of both x and $y \rightarrow w(x, y)$

$\rightarrow$ one way to determine this constant (along a streamline) is by considering the distribution of the velocity function at the inlet using the B.C

Idea $\rightarrow w$ is const along a streamline so I will use B.C to get it's value along B.C so I know it along the whole domain so the only unknown is stream function

$\rightarrow$ So I only need to substitute w with B.C in equation (4) and then solve the problem with FEM for the ψ unknown

Case (a)

$$u = u_m \quad \psi = 0 \quad \rightarrow w = \frac{\partial \psi}{\partial x} - \frac{\partial u}{\partial y} = \frac{\partial 0}{\partial x} - \frac{\partial (u_m)}{\partial y} = 0 \rightarrow w = 0$$

$$\therefore \nabla^2 \psi = 0 \quad \psi = -\int v \cdot dx = f(y) \quad \psi = \int u \cdot dy = u_m y + f(x)$$

not const
as it's double
integral

Comparing

$$\psi = u_m y \rightarrow \text{at B.C}$$

Case (b)

$$\psi = 0 \quad u = \frac{\pi u_m}{2} \cos\left(\frac{\pi y}{2d}\right) \rightarrow w = \frac{\partial \psi}{\partial x} - \frac{\partial u}{\partial y} = \frac{\partial 0}{\partial x} - \frac{\partial \left(\frac{\pi u_m}{2} \cos\left(\frac{\pi y}{2d}\right)\right)}{\partial y} = \frac{\pi^2 u_m}{4d} \sin\left(\frac{\pi y}{2d}\right)$$

$$\psi = -\int v \cdot dx = f(y) \quad \psi = \int u \cdot dy = \int \frac{\pi u_m}{2} \cos\left(\frac{\pi y}{2d}\right) \cdot dy = u_m d \sin\left(\frac{\pi y}{2d}\right) + f(x)$$

$$\psi = u_m d \sin\left(\frac{\pi y}{2d}\right) = \frac{4d^2}{\pi^2} w$$

$$\therefore \nabla^2 \psi = -\frac{\pi^2}{4d^2} \psi = k \psi$$

Final form

$$\nabla_*^2 \psi^* = C \psi^*$$

$$C = 0 \rightarrow \text{Case (a)}$$

$$= \frac{-\pi^2}{4} \rightarrow \text{Case (b)}$$

$$\text{B.C} \rightarrow \psi^*|_{inlet} = y^* \rightarrow \text{Case (a)}$$

$$= \sin \frac{\pi y^*}{2} \rightarrow \text{Case (b)}$$

$$\psi^* = 0 \rightarrow \text{at axis of symmetry}$$

$$\psi^* = 1 \rightarrow \text{on the upper wall}$$

$$\frac{\partial \psi^*}{\partial n} = 0 \rightarrow \text{at the exit}$$

$$\psi^* = \frac{\psi}{u_m d} \quad x^* = \frac{x}{d} \quad y^* = \frac{y}{d}$$

$$\nabla^2 \Psi = c \Psi$$

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} = c \Psi$$

$$R = \frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} - c \Psi$$

$$\iint_A R N_i \cdot dA = 0$$

$$\iint_A \left(\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} - c \Psi \right) N_i \cdot dA = 0$$

$$\rightarrow \iint \frac{\partial^2 \Psi}{\partial x^2} N_i \cdot dA + \iint \frac{\partial^2 \Psi}{\partial y^2} N_i \cdot dA - \iint c \Psi N_i \cdot dA = 0$$

$$\oint \Psi_x N_i \cdot dy - \iint \Psi_x N_{ix} \cdot dA + \oint \Psi_y N_i \cdot dx - \iint \Psi_y N_{iy} \cdot dA - \iint c \Psi N_i \cdot dA = 0$$

$$\rightarrow \underbrace{\oint \Psi_x N_i \cdot dy + \oint \Psi_y N_i \cdot dx}_{\text{B.C}} - \underbrace{\iint \Psi_x N_{ix} \cdot dA + \iint \Psi_y N_{iy} \cdot dA}_{\text{local Matrix}} - \iint c \Psi N_i \cdot dA = 0$$

Recall that

$$\Psi = \Psi_j N_j \rightarrow \Psi_x = \Psi_j N_{jx} \quad \Psi_y = \Psi_j N_{jy}$$

local Matrix,

$$\iint \Psi_j N_{jx} N_{ix} \cdot dA + \iint \Psi_j N_{jy} N_{iy} \cdot dA + \iint c \Psi_j N_j N_i \cdot dA = 0$$

$$\Psi_j \iint \underbrace{[N_{jx} N_{ix} + N_{jy} N_{iy} + c N_j N_i]}_{K_{ij}} \cdot dA = 0$$

$$\therefore K_{ij} \Psi_j = 0$$

$$\therefore K_{N \times N} = \iint [N_x \quad N_y]_{N \times 2} \begin{bmatrix} N_x \\ N_y \end{bmatrix}_{2 \times N} + C [N]_{N \times 1} [N]_{1 \times N} \cdot dx \cdot dy$$

$$\text{Let } \begin{bmatrix} N_x \\ N_y \end{bmatrix} = [B]$$

$$\therefore K_{N \times N} = \iint [B]_{N \times 2}^T [B]_{2 \times N} + C [N]_{N \times 1} [N]_{1 \times N} \cdot dx \cdot dy$$

X-Velocity Component

$$R = u - \frac{\partial \Psi}{\partial y}$$

$$\iint R N_i \cdot dx \cdot dy = 0$$

$$\therefore \iint u N_i \cdot dx \cdot dy = \iint \Psi_y N_i \cdot dx \cdot dy$$

$$u_j \underbrace{\iint N_j N_i \cdot dx \cdot dy}_{K_{ij}} = \Psi_j \underbrace{\iint N_{j,y} N_i \cdot dx \cdot dy}_{RHS_j}$$

$$\therefore K_{ij} u_j = RHS_j$$

$$[K]_{N \times N} = \iint [N]_{N \times 1} [N]^T_{1 \times N} \cdot dx \cdot dy$$

$$[RHS]_{N \times 1} = \iint [N]_{N \times 1} \underbrace{[N]^T_{1 \times N} \Psi}_{[M]_{N \times N}}$$

$$= [M]_{N \times N} [Y]_{N \times 1}$$

$$* [Y]_{N \times 1}$$

→ Change sign
Because of
the transpose

Y - Velocity Component

$$R = v + \frac{\partial \psi}{\partial x}$$

$$\iint R N_i \cdot dX \cdot dY = 0$$

$$\iint v N_i \cdot dX \cdot dY = - \iint \psi_x N_i \cdot dX \cdot dY$$

$$v_j \underbrace{\iint N_j N_i \cdot dX \cdot dY}_{K_{ij}} = - \psi_j \underbrace{\iint N_{xj} N_i \cdot dX \cdot dY}_{M_{ij}}$$

$$[K]_{N \times N} = \iint [N]_{N \times 1}^T [N]_{1 \times N} \cdot dX \cdot dY$$

$$[M]_{N \times N} = \iint [N]_{N \times 1}^T [N_x]_{1 \times N}$$

$$\therefore [K]_{N \times N} [v]_{N \times 1} = [M]_{N \times N} [\psi]_{N \times 1}$$

Divergence

$$R = D - u_x - v_y$$

$$\iint (D N_i - u_x N_i - v_y N_i) \cdot dx \cdot dy = 0$$

$$\iint (D_j N_j N_i - u_j N_{x_j} N_i - v_j N_{y_j} N_i) \cdot dx \cdot dy = 0$$

$$D_j \underbrace{\iint N_j N_i \cdot dx \cdot dy}_{K_{ij}} = u_j \underbrace{\iint N_{x_j} N_i \cdot dx \cdot dy}_{M_{u_{ij}}} + v_j \underbrace{\iint N_{y_j} N_i \cdot dx \cdot dy}_{M_{v_{ij}}}$$

$$[K]_{N \times N} = \iint [N]_{N \times 1} [N]_{1 \times N} \cdot dx \cdot dy$$

$$[M_u]_{N \times N} = \iint [N]_{N \times 1} [N_x]_{1 \times N} \cdot dx \cdot dy$$

$$[M_v]_{N \times N} = \iint [N]_{N \times 1} [N_y]_{1 \times N} \cdot dx \cdot dy$$

$$[K]_{N \times N} [D]_{N \times 1} = [M_v]_{N \times N} [u]_{N \times 1} + [M_u]_{N \times N} [v]_{N \times 1}$$

Pressure

$$\nabla^2 P = C_\varepsilon (u_x + v_y)$$

$$\therefore \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} = C_\varepsilon u_x + C_\varepsilon v_y$$

$$R = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} - C_\varepsilon u_x - C_\varepsilon v_y$$

$$\iint \frac{\partial^2 P}{\partial x^2} N_i \, dx \, dy + \iint \frac{\partial^2 P}{\partial y^2} N_i \, dx \, dy = C_\varepsilon \iint (u_x N_i + v_y N_i) \, dx \, dy$$

$$\begin{aligned} & \int_{BC} \frac{\partial P}{\partial x} N_i \, dy - \iint \frac{\partial P}{\partial x} N_{x,i} \, dx \, dy + \int_{BC} \frac{\partial P}{\partial y} N_i \, dx - \iint \frac{\partial P}{\partial y} N_{y,i} \, dx \, dy \\ & = C_\varepsilon \iint (u_x N_i + v_y N_i) \, dx \, dy \end{aligned}$$

$$\therefore \iint \left(\frac{\partial P}{\partial x} N_{x,i} + \frac{\partial P}{\partial y} N_{y,i} + C_\varepsilon u_x N_i + C_\varepsilon v_y N_i \right) \, dx \, dy = 0$$

$$\therefore \underbrace{\iint (N_{x,j} N_{x,i} + N_{y,j} N_{y,i}) \, dx \, dy}_{K_{ij} \equiv K_{ij}(\text{Pois}, C=0)} = -C_\varepsilon u_j \underbrace{\iint N_{x,j} N_i \, dx \, dy}_{M_{uj}} - C_\varepsilon v_j \underbrace{\iint N_{y,j} N_i \, dx \, dy}_{M_{vj}}$$

$$[K]_{N \times N} = \iint [N_x]_{N \times 1} [N_y]_{1 \times N} \begin{bmatrix} N_x \\ N_y \end{bmatrix}_{2 \times N} \, dx \, dy$$

$$[M_u]_{N \times N} = \iint [N]_{N \times 1} [N_y]_{1 \times N} \, dx \, dy$$

$$[M_v]_{N \times N} = \iint [N]_{N \times 1} [N_x]_{1 \times N} \, dx \, dy$$

$$\therefore [K]_{N \times N} [P]_{N \times 1} = -C_\varepsilon [M_v]_{N \times N} [u]_{N \times 1} - C_\varepsilon [M_u]_{N \times N} [v]_{N \times 1}$$